Test of Mathematics for University Admission

DOOMSDAY

Set 1 · Paper 1

20 questions · 75 minutes · no calculator

A paper by

Viluxent (mkb6)

Twenty original questions, each calibrated to sit beyond the difficulty of an official Question 20. Every question lies inside the TMUA Paper 1 specification. None of them can be broken open by substituting values, by working backwards from the options, or by blind computation inside the time available. Full worked solutions and an answer key follow the paper.

Before you begin

If you have opened this paper, you have already decided that a 6.5 is not what you are here for. Good. Read this before you start the clock.

This paper is deliberately harsher than the real thing. Every question was built to sit past the difficulty of an official Question 20, and every one of them has a short, clean route hidden somewhere inside it. You will not find that route by substituting numbers into the answer options; we have written them so that this does not work. You will find it by understanding the *structure* of the question. That is the entire skill the TMUA is testing.

So: 10 out of 20 on this paper is an outstanding score. Do not panic if you get 5 or 6 on your first sitting. Sit it properly: under time, no calculator, no notes. Then mark it. Then, and this is the part almost everybody skips, go back to every question you got wrong and rewrite the solution yourself from a blank page, without looking. If you cannot reproduce it, you have not learned it yet.

A word on everything else. Do the past papers, all of them, twice. Work through every free resource you can find from the people who genuinely make good TMUA material. There is far more of it now than there has ever been, and it costs you nothing but your time. And if you get stuck on something in here, or on anything else (TMUA questions, revision plans, admissions interviews, what to read), **DM us**. We would much rather answer the question than watch you lose a week to it.

One last thing.

Don't burn yourself out, but don't waste your opportunities either.

If your goal is a **9.0**, then your daily decisions need to reflect that goal. Every hour you spend scrolling, procrastinating, or putting revision off is an hour somebody else is using to get ahead. There is always someone out there working harder for the exact same university place you want. They are finishing another paper, reviewing another topic, correcting another mistake, while everyone else says they will "start tomorrow".

You do not have to be the smartest person in the room. You just have to be the person who keeps showing up.

Stay disciplined. Stay consistent. Trust the process. One day you will look back and realise that those small decisions to revise instead of procrastinate were the reason you got the score you wanted.

See you all with those **9.0s**.

Viluxent (mkb6)

Candidate instructions

READ CAREFULLY

- You have **75 minutes** to attempt all **20** questions.
- **No calculator** may be used. No formula sheet is provided or needed.
- Each question has exactly one correct answer among the options given. **The number of options varies between questions**, so read each list carefully before answering.
- There is no penalty for a wrong answer, so attempt everything.
- Only the option you select is marked; use the surrounding space for rough working.

A note on style

Every question combines at least two areas of the Paper 1 specification. Between them the twenty questions draw on indices and logarithms, algebra and polynomials, sequences, series and recurrence relations, coordinate and circle geometry, trigonometry, radian measure and sector area, differentiation, integration, functions and composite functions, and number. None of them is designed to be solved by grinding in the time available; each has a short route once the key idea is found.

The options have been chosen adversarially. In most questions the distractors are exactly the values produced by the standard mistakes: reading a construction one step wrong, taking only one branch of a case split, mis-counting a repeated root, or stopping a telescoping sum one term early. Recognising an option therefore tells you nothing.

Difficulty

This paper is aimed at candidates working towards **9.0**. Every question is calibrated to sit meaningfully beyond the difficulty of an official Question 20.

13–20	exceptional; you are ready
10–12	an outstanding result on this paper
6–9	strong; work through the solutions properly
0–5	entirely normal on a first sitting

Sources adapted for this set include the Korean CSAT, together with problems reworked from olympiad and invitational material. No question is a reproduction: every one has been restructured, re-parameterised and re-verified so that the original answer is of no use.

Question 1

The positive real numbers x and y satisfy

$$x^y = y^x \qquad \text{and} \qquad 2x = 3y.$$

Find xy.

A $\frac{9}{4}$

B $\frac{27}{8}$

C $\frac{81}{16}$

D $\frac{45}{8}$

E $\frac{243}{32}$

F $\frac{729}{64}$

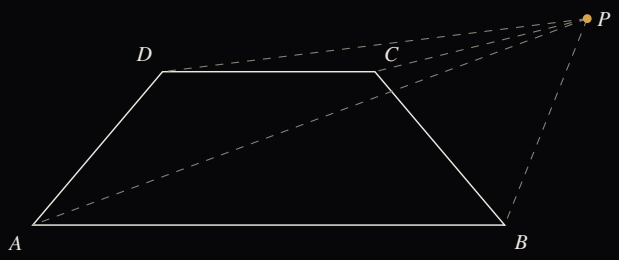
Question 2

T is an isosceles trapezium, and P is a point in its plane whose distances to the four vertices of T are

$$5, \qquad 7, \qquad 8, \qquad 12$$

in some order. Find the largest possible value of

$$\frac{\text{length of the longer parallel side of } T}{\text{length of the shorter parallel side of } T}.$$



A 12

B 8

C $\frac{119}{15}$

D $\frac{57}{11}$

E $\frac{24}{7}$

F $\frac{10}{3}$

G $\frac{95}{39}$

Question 3

The polynomial P has degree 4 and satisfies

$$P(n) = \frac{n}{n+1} \qquad \text{for } n = 0, 1, 2, 3, 4.$$

Find $P(5)$.

A 0

B $\frac{1}{2}$

C $\frac{7}{12}$

D $\frac{2}{3}$

E $\frac{5}{6}$

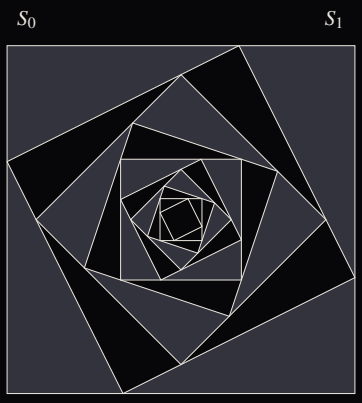
F 1

Question 4

S_0 is a square. For each $n \geq 0$ the vertices of S_{n+1} divide the sides of S_n in the ratio $1:2$ when n is even, and in the ratio $1:3$ when n is odd, every division being taken in the same cyclic direction. The regions

$$S_0 \setminus S_1, \qquad S_2 \setminus S_3, \qquad S_4 \setminus S_5, \quad \ldots$$

are shaded. What fraction of the area of S_0 is shaded?



A $\frac{4}{9}$

B $\frac{5}{9}$

C $\frac{27}{47}$

D $\frac{8}{13}$

E $\frac{23}{36}$

F $\frac{9}{14}$

G $\frac{32}{47}$

Question 5

For a positive integer a write

$$d(a) = \frac{9!}{\gcd(a, 9!)}, \qquad e(a) = \frac{8!}{\gcd(a, 8!)}.$$

How many positive integers $a \leq 2000$ are such that $d(a)$ divides $e(a)$?

A 1000

B 666

C 222

D 74

E 25

F 24

G 8

Question 6

In triangle ABC the sides opposite A, B, C are a, b, c. Given that

$$a\cos A = b\cos B, \qquad c = 13, \qquad a+b+c = 30,$$

find the largest possible area of the triangle.

A 30

B $\dfrac{13\sqrt{30}}{2}$

C $\dfrac{65}{2}$

D $6\sqrt{30}$

E $\dfrac{13\sqrt{30}}{4}$

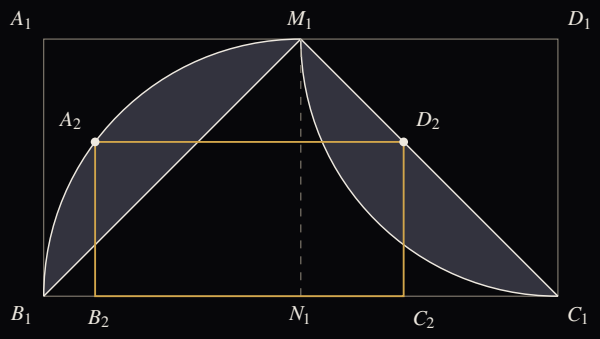
F 39

Question 7

R_1 is built as follows. In the rectangle $A_1B_1C_1D_1$ with $A_1B_1 = 1$ and $A_1D_1 = 2$, let M_1 and N_1 be the midpoints of A_1D_1 and B_1C_1. A quarter-disc sector of radius 1 is drawn with centre N_1, its arc running from B_1 to M_1; a second quarter-disc sector of radius 1 is drawn with centre D_1, its arc running from C_1 to M_1. Two regions are shaded: the region between arc M_1B_1 and chord M_1B_1, and the region between arc M_1C_1 and chord M_1C_1.

A rectangle $A_2B_2C_2D_2$ is then drawn with A_2 on the *arc* M_1B_1, D_2 on the *chord* M_1C_1, with B_2 and C_2 on B_1C_1, and with $A_2B_2 : A_2D_2 = 1 : 2$. Carrying out the whole construction inside $A_2B_2C_2D_2$ gives R_2, and so on indefinitely.

If S_n is the total shaded area of R_n, find $\lim_{n\to\infty} S_n$.



A $\dfrac{25}{24}\left(\dfrac{\pi}{2}-1\right)$

B $\dfrac{25}{21}\left(\dfrac{\pi}{2}-1\right)$

C $\dfrac{25}{16}\left(\dfrac{\pi}{2}-1\right)$

D $\dfrac{25}{9}\left(\dfrac{\pi}{2}-1\right)$

E $\dfrac{5}{4}\left(\dfrac{\pi}{2}-1\right)$

F $\dfrac{9}{5}\left(\dfrac{\pi}{2}-1\right)$

Question 8

Let a, b, c be numbers, no two of which have sum 0, such that

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}=9, \qquad \frac{a^2}{b+c}+\frac{b^2}{c+a}+\frac{c^2}{a+b}=32,$$

$$\frac{a^3}{b+c}+\frac{b^3}{c+a}+\frac{c^3}{a+b}=116.$$

Find abc.

A 1

B 2

C 4

D 6

E 12

F 13

Question 9

The line ℓ is a tangent to the curve

$$y = x^4 - 4x^3 + 2x^2 + 4x$$

at two distinct points. Find the area of the region enclosed between ℓ and the curve.

A $\dfrac{32\sqrt{2}}{15}$

B $\dfrac{8\sqrt{2}}{3}$

C $\dfrac{64}{15}$

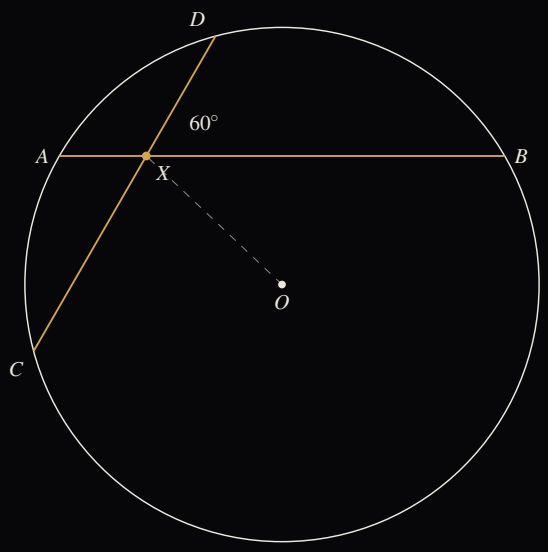
D $\dfrac{16\sqrt{2}}{5}$

E $\dfrac{128\sqrt{2}}{15}$

F $\dfrac{64\sqrt{2}}{15}$

Question 10

AB and CD are chords of a circle of radius 1, with $AB = \sqrt{3}$ and $CD = \sqrt{2}$. They cross at a point X inside the circle, the angle between them being $60°$. Find $AX \cdot XB$.



A $\dfrac{1}{3}$

B $\dfrac{\sqrt{6}}{6}$

C $\dfrac{\sqrt{3}}{3}$

D $\dfrac{\sqrt{2}}{3}$

E $\dfrac{1}{2}$

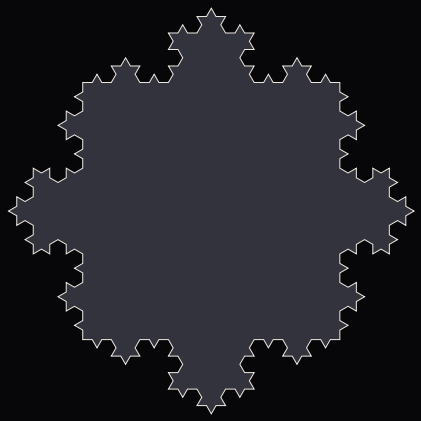
F $\dfrac{3-\sqrt{2}}{3}$

Question 11

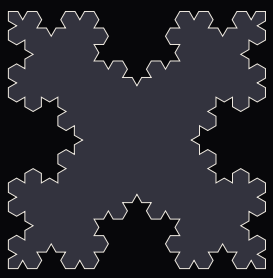
Two figures are constructed from the same square of side 3.

For the first figure, at every stage the middle third of each edge is replaced by the other two sides of an equilateral triangle erected *outwards* on it. For the second figure the triangles point *inwards* instead, and the regions they cover are removed. Both constructions continue indefinitely, and the limiting areas are A and B respectively.

Find $\dfrac{A}{B}$.



area A



area B

A $\dfrac{14+5\sqrt{3}}{11}$

B $\dfrac{14-5\sqrt{3}}{11}$

C $\dfrac{5+\sqrt{3}}{4}$

D 2

E 4

F $\dfrac{28+10\sqrt{3}}{11}$

Question 12

Let $f(x) = x^2 - 2$. How many real solutions does

$$f\big(f(f(x))\big) = x$$

have?

A 2

B 3

C 4

D 6

E 7

F 8

G 16

Question 13

The integers $a < b < c$ satisfy

$$a+b+c = 95, \qquad a^2+b^2+c^2 = 3083.$$

Find c.

A 33

B 35

C 36

D 37

E 38

F 39

G 41

Question 14

For each integer $n \geq 1$ let $f(n)$ be the number of lists of *distinct* positive integers which begin with 1, end with n, and in which every term except the last divides the term after it. For example $f(1) = 1$, $f(2) = 1$ and $f(6) = 3$, the three lists being $1, 6$ and $1, 2, 6$ and $1, 3, 6$.

How many integers n with $1 < n \leq 100$ satisfy $f(n) = 8$?

A 2

B 8

C 15

D 16

E 17

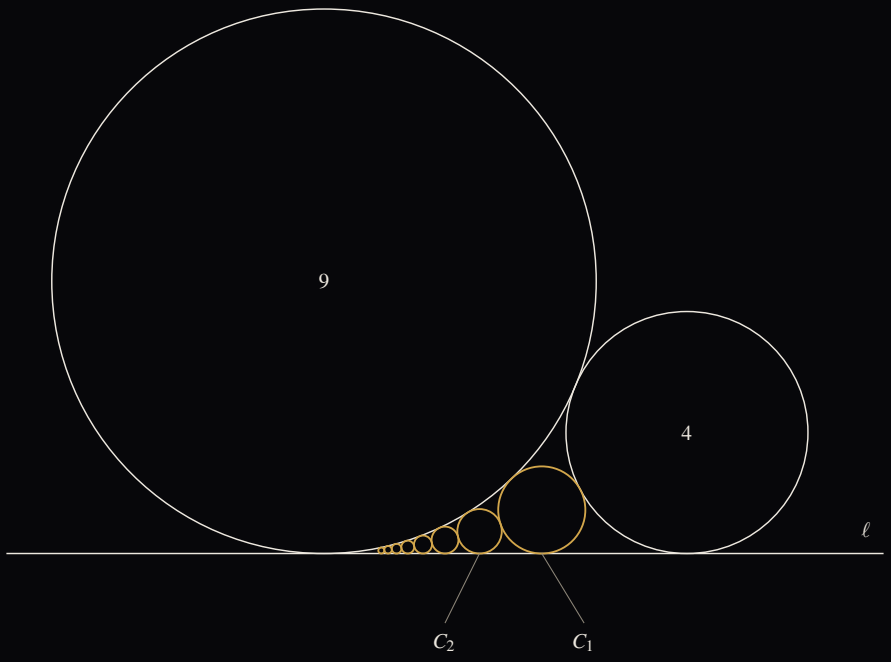
F 19

G 24

Question 15

Two circles, of radii 9 and 4, touch one another and both touch the line ℓ, on the same side of ℓ. The circle C_1 touches ℓ and both of these circles, and lies in the gap between them. For each $n \geq 1$, the circle C_{n+1} touches ℓ, touches the circle of radius 9, and touches C_n, lying between them.

How many of the circles $C_1, C_2, C_3, \ldots$ have radius greater than $\dfrac{1}{100}$?



A 26

B 27

C 28

D 29

E 30

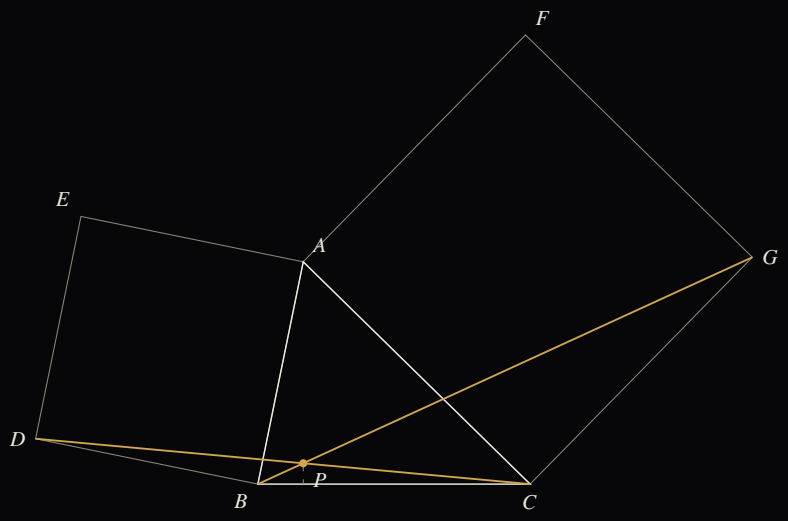
F 55

G 56

Question 16

In triangle ABC, $AB = 10$, $BC = 12$ and $CA = 14$. Squares $ABDE$ and $ACGF$, with vertices named in that cyclic order, are drawn outside the triangle. The lines BG and CD meet at P.

Find the distance from P to the line BC.



A $\dfrac{15-5\sqrt{6}}{6}$

B $\dfrac{5\sqrt{6}}{6}$

C $\dfrac{2\sqrt{6}}{3}$

D $\dfrac{5(3-\sqrt{6})}{3}$

E $\dfrac{10(3-\sqrt{6})}{3}$

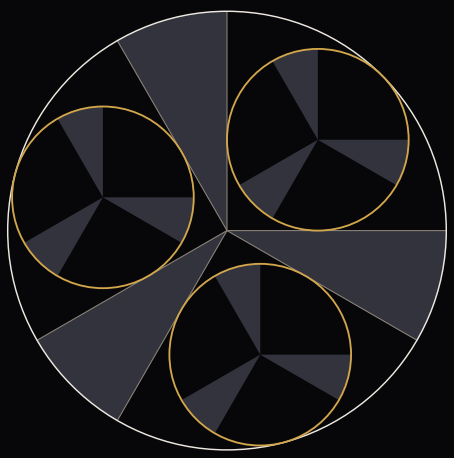
F $\dfrac{5(3+\sqrt{6})}{3}$

Question 17

Inside a circle of radius 1, three non-overlapping sectors of that circle, each of central angle 90°, are drawn; the region lying inside the circle but outside all three sectors is shaded. Call the result Figure 1.

In each of the three sectors a circle is inscribed, touching both bounding radii and the arc. Inside each of those circles the same construction is carried out, with three non-overlapping 90° sectors and the region outside them shaded, giving Figure 2; and so on indefinitely.

If S_n is the total shaded area of Figure n, find $\lim_{n\to\infty} S_n$.



A $\dfrac{\pi}{4}$

B $\dfrac{\pi(2+\sqrt{2})}{8}$

C $\dfrac{3\pi(2+\sqrt{2})}{16}$

D $\dfrac{\pi(3+2\sqrt{2})}{8}$

E $\dfrac{\pi(8+3\sqrt{2})}{16}$

F $\dfrac{\pi(4+3\sqrt{2})}{16}$

G $\dfrac{\pi(4+3\sqrt{2})}{8}$

Question 18

In triangle ABC, $BC = 6$, $CA = 5$ and $AB = 4$. A point P_0 is chosen on BC, and then for each $n \geq 0$:

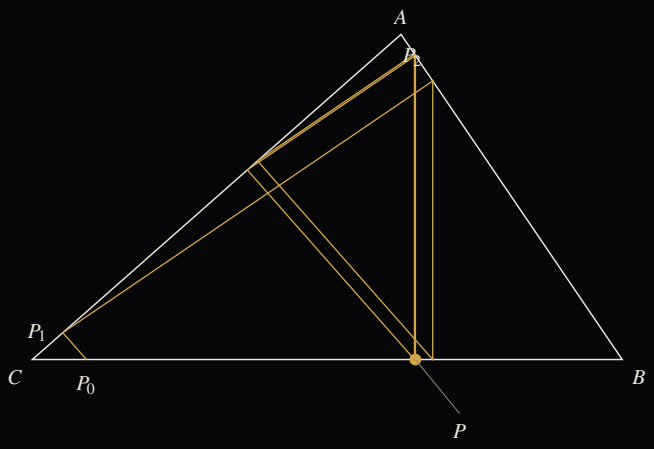
$$P_{3n+1} = \text{foot of the perpendicular from } P_{3n} \text{ to } CA,$$

$$P_{3n+2} = \text{foot of the perpendicular from } P_{3n+1} \text{ to } AB,$$

$$P_{3n+3} = \text{foot of the perpendicular from } P_{3n+2} \text{ to } BC.$$

Whatever the choice of P_0, the points $P_0, P_3, P_6, \ldots$ converge to a single point P on BC.

Find CP.



A $\dfrac{300}{77}$

B $\dfrac{162}{77}$

C $\dfrac{24}{7}$

D $\dfrac{15}{4}$

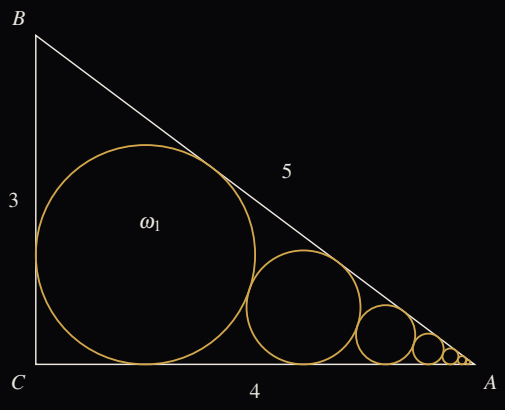
E $\dfrac{525}{128}$

F $\dfrac{100}{21}$

Question 19

A triangle has sides 3, 4 and 5, and A is the vertex at which the sides of lengths 4 and 5 meet. Let ω_1 be the incircle of the triangle, and for each $n \geq 1$ let ω_{n+1} be the circle which touches the same two sides of the triangle as ω_1 does at A, touches ω_n, and is smaller than ω_n.

Find the total area of all the circles $\omega_1, \omega_2, \omega_3, \ldots$



A $\dfrac{\pi\left(11\sqrt{10}-20\right)}{40}$

B $\dfrac{\pi\left(20+11\sqrt{10}\right)}{40}$

C $\dfrac{4\pi}{5}$

D $\dfrac{5\pi}{4}$

E $\dfrac{\pi\left(10+\sqrt{10}\right)}{9}$

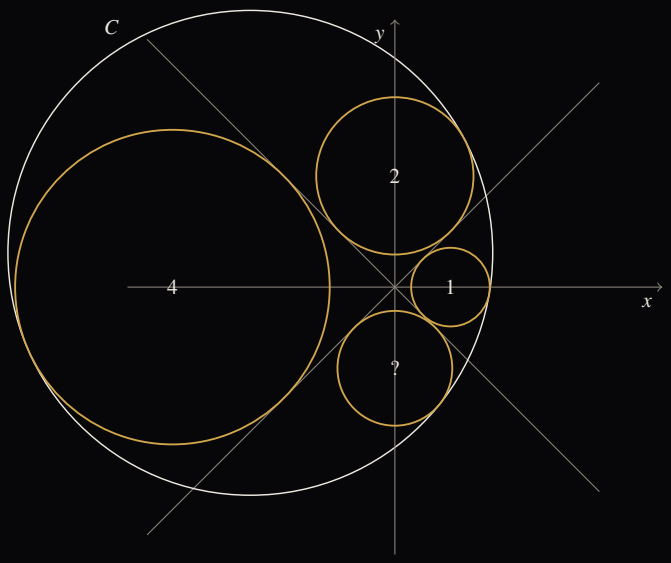
F $\dfrac{9\pi}{5}$

G $\dfrac{\pi\left(20+11\sqrt{10}\right)}{20}$

Question 20

The origin lies inside the circle C. The lines $y = x$ and $y = -x$ divide the interior of C into four regions, and each of the four half-axes lies in exactly one of these regions. In each region there is exactly one circle which touches both lines and touches C internally.

The circles in the regions containing the positive x-axis, the positive y-axis and the negative x-axis have radii 1, 2 and 4 respectively. What is the radius of the circle in the region containing the negative y-axis?



A $\dfrac{21-\sqrt{41}}{20}$

B $\dfrac{6}{5}$

C $\dfrac{9-\sqrt{41}}{2}$

D $\dfrac{3}{2}$

E $\dfrac{8}{5}$

F $\dfrac{41-\sqrt{41}}{20}$

G $\dfrac{21+\sqrt{41}}{10}$

H $\dfrac{21-\sqrt{41}}{10}$

Answer key

Mark before reading on

Q	Ans	Value	Q	Ans	Value
1	E	$\frac{243}{32}$	11	A	$\frac{14+5\sqrt{3}}{11}$
2	C	$\frac{119}{15}$	12	F	8
3	D	$\frac{2}{3}$	13	D	37
4	G	$\frac{32}{47}$	14	E	17
5	F	24	15	C	28
6	B	$\frac{13\sqrt{30}}{2}$	16	D	$\frac{5(3-\sqrt{6})}{3}$
7	C	$\frac{25}{16}\left(\frac{\pi}{2}-1\right)$	17	F	$\frac{\pi(4+3\sqrt{2})}{16}$
8	B	2	18	A	$\frac{300}{77}$
9	F	$\frac{64\sqrt{2}}{15}$	19	B	$\frac{\pi(20+11\sqrt{10})}{40}$
10	D	$\frac{\sqrt{2}}{3}$	20	H	$\frac{21-\sqrt{41}}{10}$

Mark honestly, then read the solution to every question you did not get out cleanly, including the ones you guessed correctly.

Question 1 *answer E*

Substitute $x = \frac{3}{2}y$ into $x^y = y^x$:

$$\left(\tfrac{3}{2}y\right)^y = y^{\frac{3}{2}y} \qquad \Longrightarrow \qquad \left(\tfrac{3}{2}\right)^y y^y = y^{\frac{3}{2}y} \qquad \Longrightarrow \qquad \left(\tfrac{3}{2}\right)^y = y^{\frac{1}{2}y}.$$

Both sides are positive and $y > 0$, so taking the y-th root is legitimate and gives

$$\tfrac{3}{2} = y^{1/2} \qquad \Longrightarrow \qquad y = \frac{9}{4}, \qquad x = \frac{3}{2}\cdot\frac{9}{4} = \frac{27}{8}.$$

Hence

$$xy = \frac{27}{8}\cdot\frac{9}{4} = \frac{243}{32} = \left(\frac{3}{2}\right)^5.$$

Check. $\ln(x^y) = \frac{9}{4}\ln\frac{27}{8} = \frac{27}{4}\ln\frac{3}{2}$ and $\ln(y^x) = \frac{27}{8}\ln\frac{9}{4} = \frac{27}{4}\ln\frac{3}{2}$, as required.

Remark. Taking logarithms first is equally quick: $y\ln x = x\ln y$ means $\dfrac{\ln x}{x} = \dfrac{\ln y}{y}$, and with $x = \frac{3}{2}y$ this reads $\dfrac{\ln\frac{3}{2} + \ln y}{\frac{3}{2}y} = \dfrac{\ln y}{y}$, i.e. $\ln\frac{3}{2} = \frac{1}{2}\ln y$. The distractor $\frac{45}{8}$ is $x+y$; $\frac{729}{64}$ is x^2.

Question 2 *answer C*

The mechanism. An isosceles trapezium is symmetric about the perpendicular bisector of its parallel sides, so it can be placed with vertices

$$(\pm a, 0) \quad \text{and} \quad (\pm b, h), \qquad a \neq b,\ h \neq 0.$$

Let $P = (x, y)$. The squared distances to $(a,0)$ and $(-a,0)$ are $(x \mp a)^2 + y^2$, so they differ by $4ax$; the squared distances to $(\pm b, h)$ differ by $4bx$. Dividing,

$$\frac{a}{b} = \frac{\text{difference of the two squared distances to the first parallel side}}{\text{difference of the two squared distances to the second parallel side}}.$$

So the ratio of the parallel sides is forced by nothing more than which pair of the four given distances belongs to which side. (Note $x \neq 0$, since otherwise the distances within each pair would be equal.)

The three cases. The squares are 25, 49, 64, 144, and there are three ways to split them into two pairs:

$$\begin{aligned}
\{25,49\},\{64,144\} &\to 24 \text{ and } 80, && \text{ratio } \tfrac{80}{24} = \tfrac{10}{3};\\
\{25,64\},\{49,144\} &\to 39 \text{ and } 95, && \text{ratio } \tfrac{95}{39};\\
\{25,144\},\{49,64\} &\to 119 \text{ and } 15, && \text{ratio } \tfrac{119}{15}.
\end{aligned}$$

The longer side is always the one with the larger difference, so the largest candidate is $\frac{119}{15}$.

It is attained. Take $x = 5$, so $a = \frac{119}{20}$ and $b = \frac{3}{4}$. Then $y^2 = 25 - \left(5 - \frac{119}{20}\right)^2 > 0$ and $(y-h)^2 = 49 - \left(5 - \frac{3}{4}\right)^2 > 0$ both have real solutions, giving $y \approx 4.909$ and $h \approx -0.653$. The four distances from P to

$(\pm\frac{119}{20},0)$ and $(\pm\frac{3}{4},h)$ are exactly $5,12$ and $7,8$, and $a\neq b$, $h\neq 0$, so the trapezium is genuine. Hence the answer is $\dfrac{119}{15}$.

Trap. P need not lie inside T, and in the extremal configuration it does not.

Question 3 *answer D*

Clearing the denominator is the whole idea. Put

$$Q(x)=(x+1)P(x)-x.$$

Then Q has degree 5 and $Q(n)=0$ for $n=0,1,2,3,4$, so

$$Q(x)=cx(x-1)(x-2)(x-3)(x-4)$$

for some constant c. Setting $x=-1$ kills the unknown $P(-1)$:

$$Q(-1)=0\cdot P(-1)+1=1=c(-1)(-2)(-3)(-4)(-5)=-120c \quad\Longrightarrow\quad c=-\frac{1}{120}.$$

Hence $Q(5)=c\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1=-1$, and $Q(5)=6P(5)-5$ gives

$$6P(5)-5=-1, \qquad P(5)=\frac{2}{3}.$$

Remark. The same computation gives $P(6)=0$, which is why 0 is offered.

Question 4 *answer G*

If the vertices of the new square divide the sides in the ratio $1:m$, then a side of the new square is the hypotenuse of a right triangle with legs $\frac{1}{1+m}$ and $\frac{m}{1+m}$ of the old side, so the areas are in the ratio

$$k=\frac{1+m^2}{(1+m)^2}.$$

Thus $m=2$ gives $k_1=\frac{5}{9}$ and $m=3$ gives $k_2=\frac{10}{16}=\frac{5}{8}$.

Take $[S_0]=1$. Then $[S_1]=k_1$, $[S_2]=k_1k_2$, $[S_3]=k_1^2k_2$, and in general $[S_{2j}]=(k_1k_2)^j$. The shaded area is

$$\sum_{j\geq 0}\Big([S_{2j}]-[S_{2j+1}]\Big)=\sum_{j\geq 0}(k_1k_2)^j(1-k_1)=\frac{1-k_1}{1-k_1k_2}=\frac{\frac{4}{9}}{1-\frac{25}{72}}=\frac{\frac{4}{9}}{\frac{47}{72}}=\frac{32}{47}.$$

Trap. With a constant ratio $1:2$ throughout the answer would be $\frac{1-k_1}{1-k_1^2}=\frac{1}{1+k_1}=\frac{9}{14}$.

Question 5 *answer F*

Divisibility can be checked one prime at a time. For a prime p, writing v_p for the exponent of p,

$$v_p\big(d(a)\big)=v_p(9!)-\min\big(v_p(a),v_p(9!)\big),$$

and likewise for $e(a)$ with 8! in place of 9!. So $d(a) \mid e(a)$ exactly when

$$v_p(9!) - \min\big(v_p(a), v_p(9!)\big) \;\le\; v_p(8!) - \min\big(v_p(a), v_p(8!)\big) \qquad \text{for every prime } p.$$

Only one prime matters. Since $9! = 9 \cdot 8!$ and $9 = 3^2$, the exponents of 2, 5 and 7 are the same in 9! as in 8!, so for those primes the two sides are identical and the condition holds automatically. For $p = 3$,

$$v_3(9!) = \left\lfloor \tfrac{9}{3} \right\rfloor + \left\lfloor \tfrac{9}{9} \right\rfloor = 4, \qquad v_3(8!) = \left\lfloor \tfrac{8}{3} \right\rfloor + \left\lfloor \tfrac{8}{9} \right\rfloor = 2.$$

The condition. Put $v = v_3(a)$. The requirement is $4 - \min(v,4) \le 2 - \min(v,2)$:

$$\begin{array}{lll} v \le 2: & 4 - v \le 2 - v & \text{false;} \\ v = 3: & 1 \le 0 & \text{false;} \\ v \ge 4: & 0 \le 0 & \text{true.} \end{array}$$

So the condition is precisely $3^4 = 81 \mid a$, and the count is

$$\left\lfloor \frac{2000}{81} \right\rfloor = 24$$

(as $81 \cdot 24 = 1944$ and $81 \cdot 25 = 2025$).

Traps. Requiring only $27 \mid a$ gives 74; requiring only $3 \mid a$ gives 666; requiring $243 \mid a$ gives 8.

Question 6 *answer B*

Substitute the cosine rule into the condition:

$$a \cdot \frac{b^2 + c^2 - a^2}{2bc} = b \cdot \frac{a^2 + c^2 - b^2}{2ac} \implies a^2(b^2 + c^2 - a^2) = b^2(a^2 + c^2 - b^2),$$

so $c^2(a^2 - b^2) = a^4 - b^4$ and therefore

$$(a^2 - b^2)\big(c^2 - a^2 - b^2\big) = 0.$$

Hence $a = b$ **or** $C = 90^\circ$, and both branches must be examined.

$C = 90^\circ$**.** Then $a + b = 17$ and $a^2 + b^2 = 169$, so $2ab = 289 - 169 = 120$ and the area is $\frac{1}{2}ab = 30$. (Indeed $\{a,b\} = \{5,12\}$.)

$a = b$**.** Then $a = b = \frac{17}{2}$, and the height to c is $\sqrt{\left(\frac{17}{2}\right)^2 - \left(\frac{13}{2}\right)^2} = \sqrt{30}$, giving area $\frac{1}{2} \cdot 13 \cdot \sqrt{30} = \frac{13\sqrt{30}}{2}$.

Finally $\left(\frac{13\sqrt{30}}{2}\right)^2 = \frac{5070}{4} = 1267.5 > 900 = 30^2$, so the larger area is $\dfrac{13\sqrt{30}}{2}$.

Question 7 *answer C*

The shaded area of one figure. Each shaded piece is a quarter disc of radius 1 minus a right isosceles triangle of legs 1, so

$$S_1 = 2\left(\frac{\pi}{4} - \frac{1}{2}\right) = \frac{\pi}{2} - 1.$$

The scale factor. Take $B_1=(0,0)$, $C_1=(2,0)$, $A_1=(0,1)$, $D_1=(2,1)$, so $M_1=(1,1)$ and $N_1=(1,0)$. Arc M_1B_1 is part of $(x-1)^2+y^2=1$; chord M_1C_1 is part of $x+y=2$.

Let $A_2=(x,y)$. Then $B_2=(x,0)$, the height is y, the width is $2y$, and $D_2=(x+2y,\,y)$. The two conditions are

$$(x-1)^2+y^2=1, \qquad (x+2y)+y=2.$$

From the second, $x=2-3y$; substituting, $(1-3y)^2+y^2=1$, i.e. $10y^2=6y$, so $y=\frac{3}{5}$.

The child rectangle therefore has linear scale $\frac{3}{5}$ and area scale $\frac{9}{25}$, so the shaded areas form a geometric series with ratio $\frac{9}{25}$:

$$\lim_{n\to\infty} S_n=\frac{S_1}{1-\frac{9}{25}}=\frac{25}{16}\left(\frac{\pi}{2}-1\right).$$

Trap. If instead A_2 is on the chord and D_2 on the arc, which is the standard textbook configuration, the scale is $\frac{2}{5}$ and the answer is $\frac{25}{21}\left(\frac{\pi}{2}-1\right)$. Read the placement carefully.

Question 8 *answer B*

Write $s=a+b+c$, $p=ab+bc+ca$, $q=abc$, and let $T_k=\sum\dfrac{a^k}{b+c}$.

The ladder. Since $a=s-(b+c)$,

$$\frac{a^{k+1}}{b+c}=a^k\cdot\frac{a}{b+c}=a^k\cdot\frac{s-(b+c)}{b+c}=\frac{s\,a^k}{b+c}-a^k,$$

so summing cyclically,

$$\boxed{T_{k+1}=s\,T_k-\sum a^k}$$

Applying it twice:

$$T_2=sT_1-s=9s-s=8s=32\implies s=4,$$

$$T_3=sT_2-\sum a^2=4\cdot 32-\sum a^2=116\implies \sum a^2=12.$$

Hence $p=\frac{s^2-\sum a^2}{2}=\frac{16-12}{2}=2$.

Extracting q. With $k=0$ the ladder reads $T_1=s\,T_0-3$, where $T_0=\sum\frac{1}{b+c}=\sum\frac{1}{s-a}$; thus $T_0=\frac{9+3}{4}=3$. Now

$$\sum\frac{1}{s-a}=\frac{\sum(s-b)(s-c)}{(s-a)(s-b)(s-c)}=\frac{3s^2-2s\cdot s+p}{s^3-s\cdot s^2+sp-q}=\frac{s^2+p}{sp-q},$$

so

$$3=\frac{16+2}{8-q}\quad\implies\quad 8-q=6\quad\implies\quad q=2.$$

Only symmetric-function manipulation has been used, so nothing depends on the numbers being real.

Question 9 *answer F*

If $\ell:\ y=mx+c$ touches the quartic at two points, then $x^4-4x^3+2x^2+4x-mx-c$ is a quartic with two double roots and leading coefficient 1, i.e.

$$x^4-4x^3+2x^2+(4-m)x-c=(x^2+px+q)^2.$$

Comparing coefficients:

$$2p=-4\Rightarrow p=-2;\quad p^2+2q=2\Rightarrow q=-1;\quad 2pq=4-m\Rightarrow m=0;\quad q^2=-c\Rightarrow c=-1.$$

So ℓ is the horizontal line $y=-1$, and

$$\left(x^4-4x^3+2x^2+4x\right)-(-1)=(x^2-2x-1)^2\ \geq 0,$$

with equality exactly at $x=1\pm\sqrt{2}$, which confirms the double contact.

Substituting $u=x-1$ (so $x^2-2x-1=u^2-2$),

$$\int_{1-\sqrt{2}}^{1+\sqrt{2}}(x^2-2x-1)^2\,dx=\int_{-\sqrt{2}}^{\sqrt{2}}\left(u^4-4u^2+4\right)du=2\left[\frac{u^5}{5}-\frac{4u^3}{3}+4u\right]_0^{\sqrt{2}}.$$

Since $(\sqrt{2})^5=4\sqrt{2}$ and $(\sqrt{2})^3=2\sqrt{2}$, the bracket is $\sqrt{2}\left(\frac{4}{5}-\frac{8}{3}+4\right)=\frac{32\sqrt{2}}{15}$, and the area is $\dfrac{64\sqrt{2}}{15}$.

Question 10 *answer D*

Let O be the centre. A chord of length L in a unit circle lies at distance $\sqrt{1-L^2/4}$ from O, so

$$d(O,AB)=\sqrt{1-\tfrac{3}{4}}=\tfrac{1}{2},\qquad d(O,CD)=\sqrt{1-\tfrac{1}{2}}=\tfrac{\sqrt{2}}{2}.$$

Put O at the origin with AB along $y=\frac{1}{2}$. The line CD makes 60° with AB and is at distance $\frac{\sqrt{2}}{2}$ from O, so it is

$$\sqrt{3}x-y=\pm\sqrt{2}.$$

At $y=\frac{1}{2}$ this gives $x=\dfrac{\frac{1}{2}\pm\sqrt{2}}{\sqrt{3}}$, and hence

$$OX^2=x^2+\tfrac{1}{4}=\frac{\left(\frac{1}{2}\pm\sqrt{2}\right)^2}{3}+\frac{1}{4}=1\mp\frac{\sqrt{2}}{3}.$$

The upper sign gives $OX^2=1+\frac{\sqrt{2}}{3}>1$, so X would lie outside the circle; the chords could not then cross inside. Hence $OX^2=1-\dfrac{\sqrt{2}}{3}$.

By the intersecting chords theorem (the power of the point X),

$$AX\cdot XB=1-OX^2=\frac{\sqrt{2}}{3}.$$

Note. The same value is obtained for $CX\cdot XD$, as it must be.

Question 11 *answer A*

Let the square have side s, so its area is s^2.

Stage 1. Each of the 4 edges gains an equilateral triangle of side $s/3$, of area $\frac{\sqrt{3}}{4}\cdot\frac{s^2}{9}$; the total is $\frac{\sqrt{3}s^2}{9}$.

The ratio between stages. Every edge is replaced by 4 edges of one third the length, so at each stage the number of triangles is multiplied by 4 and each has $\frac{1}{9}$ of the previous area: successive stages add $\frac{4}{9}$ of the previous addition. Hence

$$\text{total added} = \frac{\sqrt{3}s^2}{9} \cdot \frac{1}{1-\frac{4}{9}} = \frac{\sqrt{3}s^2}{9} \cdot \frac{9}{5} = \frac{\sqrt{3}}{5}s^2.$$

The inward construction removes exactly the same total. Therefore

$$A = s^2\left(1+\frac{\sqrt{3}}{5}\right), \qquad B = s^2\left(1-\frac{\sqrt{3}}{5}\right),$$

and the side length 3 cancels:

$$\frac{A}{B} = \frac{5+\sqrt{3}}{5-\sqrt{3}} = \frac{(5+\sqrt{3})^2}{22} = \frac{28+10\sqrt{3}}{22} = \frac{14+5\sqrt{3}}{11}.$$

Trap. For the same construction on an equilateral triangle the answer is exactly 4; on the square it is not, because the added area is no longer a fixed fraction of the same shape.

Question 12 *answer F*

$f(f(f(x))) - x$ has degree 8, so there are at most 8 real solutions.

Nothing outside $[-2,2]$**.** If $x > 2$ then $x^2 - 2 - x = (x-2)(x+1) > 0$, so $f(x) > x > 2$ and the iterates increase; if $x < -2$ then $f(x) = x^2 - 2 > 2 > x$ and the same applies. Either way $f^3(x) > x$.

Inside $[-2,2]$**.** Put $x = 2\cos\theta$ with $\theta \in [0,\pi]$. Then

$$f(2\cos\theta) = 4\cos^2\theta - 2 = 2\cos 2\theta,$$

so $f^3(x) = 2\cos 8\theta$ and the equation becomes $\cos 8\theta = \cos\theta$, i.e.

$$8\theta = \theta + 2k\pi \quad \text{or} \quad 8\theta = -\theta + 2k\pi,$$

that is $\theta = \dfrac{2k\pi}{7}$ or $\theta = \dfrac{2k\pi}{9}$.

In $[0,\pi]$ the first family gives $\theta = 0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$ (four values) and the second gives $\theta = 0, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{6\pi}{9}, \frac{8\pi}{9}$ (five values), overlapping only at $\theta = 0$. That is $4+5-1 = 8$ values of θ, and cos is injective on $[0,\pi]$, so there are 8 distinct real solutions, the maximum possible.

(Explicitly, $f^3(x) - x = (x-2)(x+1)(x^3-3x+1)(x^3+x^2-2x-1)$.)

Question 13 *answer D*

Shifting to the mean removes the linear condition. Put

$$x = 3a - 95, \qquad y = 3b - 95, \qquad z = 3c - 95.$$

Then $x+y+z = 3\cdot 95 - 3\cdot 95 = 0$ and

$$x^2+y^2+z^2 = 9\sum a^2 - 6\cdot 95\sum a + 3\cdot 95^2 = 9(3083) - 6(95)(95) + 3(9025) = 672.$$

Eliminating $z=-(x+y)$,

$$x^2+y^2+(x+y)^2=672 \quad\Longrightarrow\quad x^2+xy+y^2=336.$$

Since $x^2+xy+y^2=\left(x+\frac{y}{2}\right)^2+\frac{3}{4}y^2$, we need $|y|\le\sqrt{448}<22$, and a short search gives (up to order and sign) only

$$\{x,y,z\}=\{16,4,-20\} \quad\text{or}\quad \{-16,-4,20\}.$$

Finally $x=3a-95\equiv 1 \pmod 3$, and likewise for y and z; the second set has residues $2 \pmod 3$ and is rejected. Hence $\{x,y,z\}=\{-20,4,16\}$ and

$$\{a,b,c\}=\left\{\frac{95-20}{3},\frac{95+4}{3},\frac{95+16}{3}\right\}=\{25,33,37\},$$

so $c=37$. (Check: $25+33+37=95$ and $625+1089+1369=3083$.)

Question 14 *answer E*

Splitting a list at its penultimate term gives the recurrence

$$f(1)=1, \qquad f(n)=\sum_{d\mid n,\ d<n} f(d).$$

Because a divisor is determined by its exponents, $f(n)$ depends only on the multiset of exponents in the prime factorisation of n. Computing the shapes that can occur for $n\le 100$:

shape	p	p^2	p^3	p^4	p^5	p^6	pq	p^2q	p^3q	p^2q^2	pqr
$f(n)$	1	2	4	8	16	32	3	8	20	26	13

(For instance $f(p^k)=2^{k-1}$, since $1+1+2+\cdots+2^{k-2}=2^{k-1}$; and $f(p^2q)=f(1)+f(p)+f(p^2)+f(q)+f(pq)=1+1+2+1+3=8$.)

So $f(n)=8$ exactly when $n=p^4$ or $n=p^2q$ with $p\neq q$ prime. Counting up to 100:

- p^4: 16 and 81 — 2 values;
- p^2q: $p=2$ needs $q\le 25$, giving $q\in\{3,5,7,11,13,17,19,23\}$ (8); $p=3$ needs $q\le 11$, giving $q\in\{2,5,7,11\}$ (4); $p=5$ gives $q\in\{2,3\}$ (2); $p=7$ gives $q=2$ (1) — 15 values.

Total: $2+15=\mathbf{17}$.

Question 15 *answer C*

A lemma. If circles of radii u and v touch ℓ on the same side and touch each other, the distance between their points of contact is

$$\sqrt{(u+v)^2-(u-v)^2}=2\sqrt{uv}.$$

If a circle of radius w sits in the gap and touches all three, the two smaller gaps make up the larger one:

$$2\sqrt{uw}+2\sqrt{wv}=2\sqrt{uv} \qquad\Longrightarrow\qquad \frac{1}{\sqrt{w}}=\frac{1}{\sqrt{u}}+\frac{1}{\sqrt{v}}.$$

The chain. Writing $r_0 = 4$ for the small starting circle and $u = 9$ throughout,

$$\frac{1}{\sqrt{r_{n+1}}} = \frac{1}{3} + \frac{1}{\sqrt{r_n}}, \qquad \frac{1}{\sqrt{r_0}} = \frac{1}{2},$$

an arithmetic progression in $1/\sqrt{r_n}$:

$$\frac{1}{\sqrt{r_n}} = \frac{1}{2} + \frac{n}{3} = \frac{2n+3}{6}, \qquad \text{so} \qquad r_n = \frac{36}{(2n+3)^2}.$$

(Check: $r_1 = \frac{36}{25}$, $r_2 = \frac{36}{49}$.) Finally

$$\frac{36}{(2n+3)^2} > \frac{1}{100} \iff (2n+3)^2 < 3600 \iff 2n+3 < 60 \iff n \le 28.$$

So 28 of the circles qualify.

Question 16 *answer D*

Put $B = (0,0)$ and $C = (a,0)$ with $A = (p,h)$. Then

$$p^2 + h^2 = 100, \qquad (p-12)^2 + h^2 = 196 \implies -24p + 144 = 96,$$

so $p = 2$, $h = \sqrt{96} = 4\sqrt{6}$ and $a = 12$.

The two square vertices. D is obtained by rotating $\overrightarrow{AB}$ about B through -90°, and G by rotating $\overrightarrow{AC}$ about C through $+90^\circ$ (this is what "outside" means for the given vertex order):

$$D = (-h,\, p), \qquad G = (a+h,\, a-p).$$

The intersection. BG is $y = \dfrac{(a-p)x}{a+h}$, and CD is $y = \dfrac{p(a-x)}{a+h}$. Equating,

$$x(a-p) = p(a-x) \implies xa = pa \implies x = p.$$

So P always lies on the altitude from A. That is the key structural fact, and it puts P at height

$$\frac{p(a-p)}{a+h} = \frac{2 \cdot 10}{12+4\sqrt{6}} = \frac{5}{3+\sqrt{6}} = \frac{5(3-\sqrt{6})}{9-6} = \frac{5(3-\sqrt{6})}{3}.$$

Question 17 *answer F*

Figure 1. Three sectors of 90° use up 270°, so

$$S_1 = \pi - 3 \cdot \frac{\pi}{4} = \frac{\pi}{4}$$

(the shading is three gaps of 30°).

The inscribed circle. In a 90° sector of the unit circle the half-angle is 45°. If the inscribed circle has radius r and centre at distance d along the bisector then $r = d\sin 45^\circ$ and $d + r = 1$, so

$$r = (1-r)\frac{\sqrt{2}}{2} \quad \implies \quad r = \frac{\sqrt{2}}{2+\sqrt{2}} = \sqrt{2} - 1.$$

The ratio. Each of the three sub-figures is a copy scaled by $\sqrt{2}-1$, so it carries $(\sqrt{2}-1)^2$ of the shading; the common ratio is

$$k = 3(\sqrt{2}-1)^2 = 3(3-2\sqrt{2}) = 9-6\sqrt{2} \approx 0.515 < 1.$$

Hence

$$\lim_{n\to\infty} S_n = \frac{\pi/4}{1-(9-6\sqrt{2})} = \frac{\pi}{4(6\sqrt{2}-8)} = \frac{\pi}{24\sqrt{2}-32} = \frac{\pi(24\sqrt{2}+32)}{1152-1024} = \frac{\pi(4+3\sqrt{2})}{16}.$$

Question 18 *answer A*

The three cosines. By the cosine rule,

$$\cos A = \frac{25+16-36}{40} = \frac{1}{8}, \qquad \cos B = \frac{36+16-25}{48} = \frac{9}{16}, \qquad \cos C = \frac{36+25-16}{60} = \frac{3}{4}.$$

All are positive, so the triangle is acute and every foot lands inside the relevant side.

One round trip. If a point of CB is at distance x from C, its foot on CA is at distance $x\cos C$ from C (project onto CA). That point is at distance $b - x\cos C$ from A, so its foot on AB is at distance $(b - x\cos C)\cos A$ from A, i.e. at distance $c - (b - x\cos C)\cos A$ from B; its foot on BC is therefore at distance $\big(c - (b - x\cos C)\cos A\big)\cos B$ from B. Writing $x_n = CP_{3n}$,

$$x_{n+1} = a - \cos B\Big(c - \cos A\big(b - x_n\cos C\big)\Big) = \Big(a - c\cos B + b\cos A\cos B\Big) - x_n\cos A\cos B\cos C.$$

With the values above,

$$x_{n+1} = \frac{525}{128} - \frac{27}{512}x_n.$$

The limit. The multiplier has modulus $\frac{27}{512} < 1$, so the sequence converges (alternating about the limit) to the fixed point:

$$x^* = \frac{525/128}{1+\frac{27}{512}} = \frac{525}{128}\cdot\frac{512}{539} = \frac{2100}{539} = \frac{300}{77}.$$

Trap. The limiting triangle is *not* the orthic triangle: the foot of the altitude from A is at distance $b\cos C = \frac{15}{4}$ from C, which is offered as an option. And $\frac{162}{77}$ is BP.

Question 19 *answer B*

The incircle. The triangle is right-angled with area 6 and semiperimeter $s = 6$, so its inradius is $r_1 = \dfrac{\text{area}}{s} = 1$. The tangent length from a vertex is s minus the opposite side, and the side opposite A is 3, so the incircle touches the two sides through A at distance $6-3 = 3$ from A. Hence, with 2α the angle at A,

$$\tan\alpha = \frac{1}{3} \quad\Longrightarrow\quad \sin\alpha = \frac{1}{\sqrt{10}}.$$

The ratio. Two circles inscribed in the same angle, touching each other, have centres at distances $d = \dfrac{r}{\sin\alpha}$ and $d' = \dfrac{r'}{\sin\alpha}$ from A with $d - d' = r + r'$, so

$$\frac{r}{\sin\alpha} - \frac{r'}{\sin\alpha} = r + r' \quad\Longrightarrow\quad k = \frac{r'}{r} = \frac{1-\sin\alpha}{1+\sin\alpha} = \frac{\sqrt{10}-1}{\sqrt{10}+1}.$$

The sum. The radii are $1, k, k^2, \ldots$, so the areas form a geometric series of ratio k^2:

$$\sum_{n\ge 1} \pi r_n^2 = \frac{\pi}{1-k^2}.$$

Now $k^2 = \dfrac{11-2\sqrt{10}}{11+2\sqrt{10}}$, so

$$1-k^2 = \frac{4\sqrt{10}}{11+2\sqrt{10}}, \qquad \frac{\pi}{1-k^2} = \frac{\pi\left(11+2\sqrt{10}\right)}{4\sqrt{10}} = \frac{\pi\left(11\sqrt{10}+20\right)}{40}.$$

Question 20 *answer H*

Setting up. Each region is a right-angled wedge with apex O, so a circle of radius r touching both lines has its centre at distance $r\sqrt{2}$ from O along the bisector, i.e. on a half-axis. Let C have centre (α,β) and radius R, and put

$$K = R^2 - \alpha^2 - \beta^2 \quad (>0, \text{ since } O \text{ is inside } C).$$

For the circle of radius $r_1 = 1$ on the positive x-axis, internal tangency gives $(\alpha - r_1\sqrt{2})^2 + \beta^2 = (R - r_1)^2$, which expands to

$$r_1^2 + 2r_1\left(R - \sqrt{2}\,\alpha\right) = K,$$

and dividing by $2r_1$,

$$R - \sqrt{2}\,\alpha = \frac{K}{2r_1} - \frac{r_1}{2}.$$

The other three circles give the same identity with $\left(R+\sqrt{2}\alpha, r_3\right)$, $\left(R-\sqrt{2}\beta, r_2\right)$ and $\left(R+\sqrt{2}\beta, r_4\right)$.

Adding opposite pairs eliminates α and β:

$$4R = K\left(\frac{1}{r_1}+\frac{1}{r_3}\right) - (r_1+r_3) = K\left(\frac{1}{r_2}+\frac{1}{r_4}\right) - (r_2+r_4).$$

With $r_1 = 1,\ r_3 = 4$ the first expression gives $4R = \frac{5K}{4} - 5$, i.e.

$$R = \frac{5K}{16} - \frac{5}{4}.$$

Closing the system. Subtracting the same pair gives $2\sqrt{2}\,\alpha = \left(\frac{K}{8}-2\right) - \left(\frac{K}{2}-\frac{1}{2}\right) = -\frac{3K}{8} - \frac{3}{2}$, and from $r_2 = 2$, $\sqrt{2}\,\beta = R - \frac{K}{4} + 1$. Substituting both into $K = R^2 - \alpha^2 - \beta^2$ leaves a single equation in K, whose relevant root is

$$K = \frac{61+9\sqrt{41}}{5}, \qquad \text{whence} \qquad R = \frac{41+9\sqrt{41}}{16} \approx 6.164.$$

The fourth radius. From $R + \sqrt{2}\beta = \frac{K}{2r_4} - \frac{r_4}{2}$ and $\sqrt{2}\beta = R - \frac{K}{4} + 1$,

$$r_4^2 + \left(4R - \frac{K}{2} + 2\right) r_4 - K = 0, \qquad 4R - \frac{K}{2} + 2 = \frac{123+27\sqrt{41}}{20},$$

and the positive root is

$$r_4 = \frac{21-\sqrt{41}}{10} \approx 1.4597.$$

Check. With $\alpha \approx -3.676$, $\beta \approx 0.872$, $R \approx 6.164$ all four tangency conditions hold to full precision, and $\sqrt{\alpha^2+\beta^2} \approx 3.778 < R$, so the origin is indeed inside C.

Provenance and syllabus notes

Two of the twenty questions are rebuilt from the Korean CSAT *deungbigeupsu dohyeong* type, a self-similar construction whose areas form a geometric progression (Questions 7 and 17). That type maps exactly onto the TMUA's sum-to-infinity and sector-area content, and is almost entirely absent from UK preparation material. Questions 4 and 11 are the same idea carried onto a square and onto a Koch-type boundary; Questions 15 and 19 replace the geometric progression with a genuine recurrence.

Questions 2, 5, 8 and 13 are rebuilt from invitational and olympiad material. In each case the original was outside the specification, or answerable by machine, or both. What has been kept is the mechanism: the difference of two squared distances that fixes the ratio of the parallel sides of an isosceles trapezium, the single prime whose exponent separates 9! from 8!, the ladder identity between $\sum a^k/(b+c)$ and $\sum a^{k+1}/(b+c)$, and the shift to the mean that turns a pair of symmetric conditions into $x^2+xy+y^2=336$. Everything else has been rebuilt around them.

On syllabus

Every question sits inside the TMUA Paper 1 specification. The following were checked explicitly:

- **No limits are examined.** Where a limit appears (Questions 7, 11, 17, 18) it is the sum of a geometric series or the fixed point of a contraction, both of which are on specification. No small-angle behaviour, no differentiation of $\ln x$ or e^x, no volumes of revolution.
- **No probability, no statistics, no complex arithmetic.** Question 8 is pure symmetric-function manipulation and never needs the numbers to be real.
- **Calculus is restricted to polynomials** (Question 9: tangency conditions and a polynomial integral).
- **Trigonometry** is limited to the sine and cosine rules, the area formula, radian measure and sector area, and exact values at standard angles. No compound-angle or double-angle identity is required anywhere: Question 6 derives its case split from the cosine rule alone.

On the answer options

Every option list was built after the solution, from the wrong answers the question actually produces. Working backwards from the options, substituting convenient values, or eliminating by rough size will not identify the correct answer in any question on this paper. Several questions offer, as a distractor, the answer to the *well-known* problem the question resembles: Question 7 offers $\frac{25}{21}\left(\frac{\pi}{2}-1\right)$, Question 11 offers 4, and Question 18 offers the foot of the altitude. Candidates who recognise a question and stop reading will find that this is punished rather than rewarded.

Viluxent mkb6

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